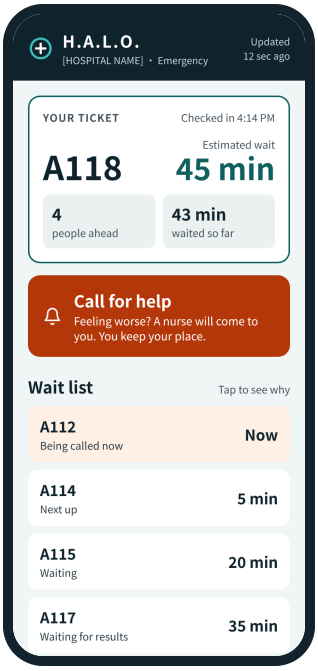
H.A.L.O.

One live wait screen for the whole waiting room, mirrored on every patient's phone

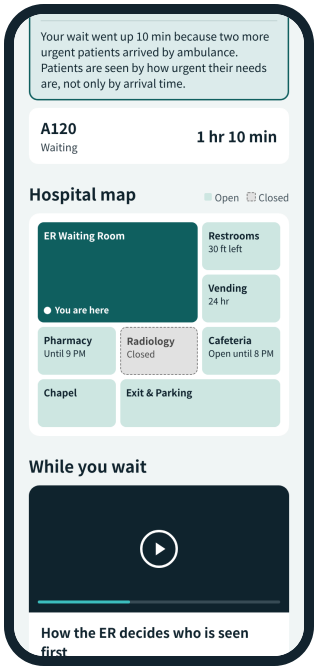
We combined our three concepts into one bigger concept system. The H.A.L.O. kiosk tower is the core, because it works for every patient, phone or no phone. From our other two concepts we added a phone page with the **same information**, opened from a QR code, and a **Call for help** button that alerts a nurse.



Kiosk tower: one of four identical screens



Phone: top of page



Phone: scrolled down

Same information, two sizes

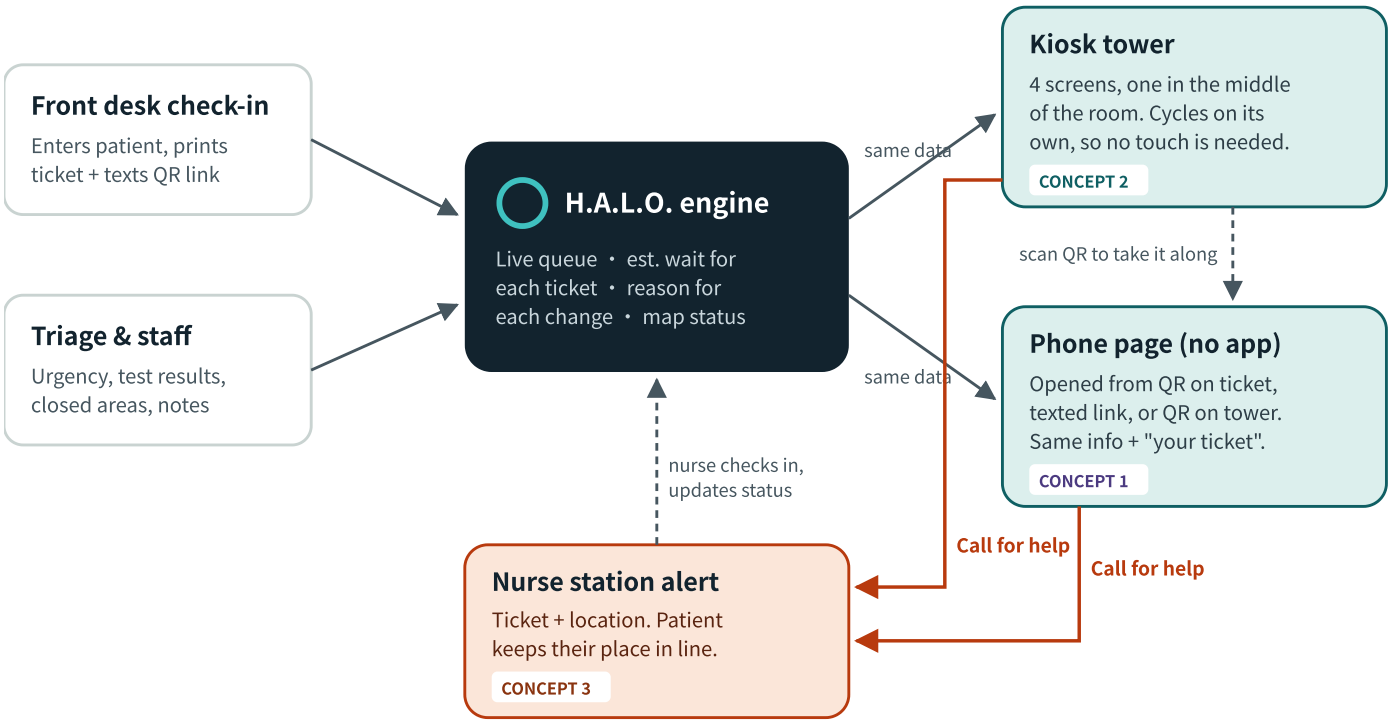
Both screens show the same four things: the **wait list** with reasons, the **hospital map**, **Call for help**, and **helpful videos**. The phone puts the patient's own ticket at the top, and the tower shows everyone's.

CONCEPT 2 H.A.L.O. kiosk: core
CONCEPT 1 QR check-in ticket: phone access
CONCEPT 3 Scan-from-seat queue: nurse alert, people ahead

HOW THE CONCEPTS WORK TOGETHER

The concept system

One live wait engine feeds every screen, so the tower and the phone can never disagree. Patients can get help from either one, and staff only have to update information in one place.



A patient's visit, step by step

1

Check in

The front desk checks the patient in as usual. They get a ticket (A118) with a QR code, and the same link is texted as a backup.

2

See the wait

They find their number on the tower, or open their phone: *45 min, 4 people ahead.*

3

Learn why

Their wait jumped 10 minutes. They tap their row and see that two more urgent patients arrived by ambulance.

4

Wait comfortably

The map shows the restrooms 30 ft away and that the cafeteria is open until 8. Videos explain how triage works.

5

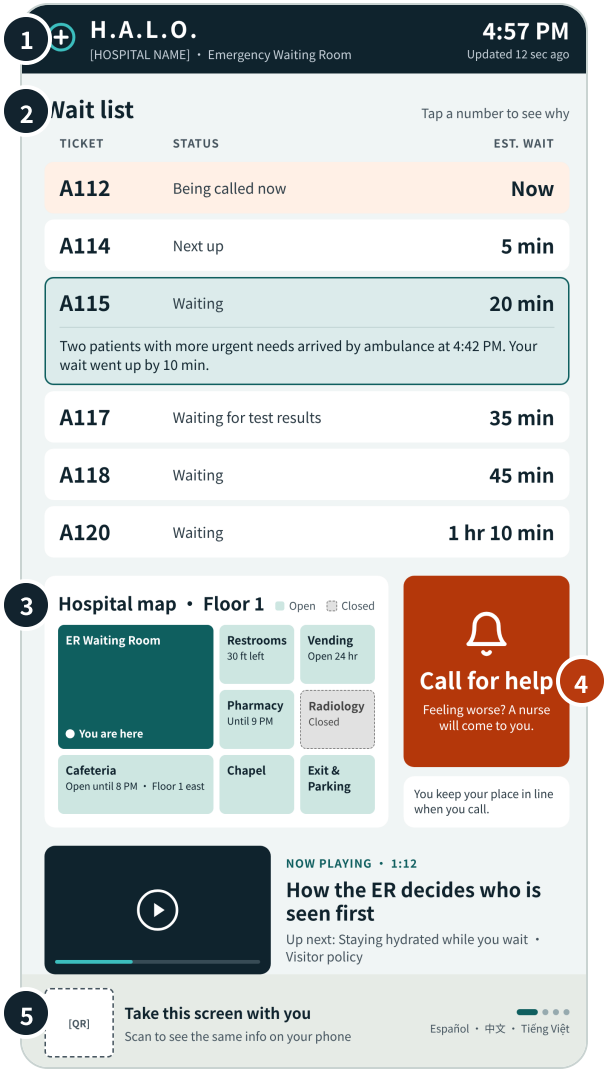
Feeling worse?

They press Call for help on the tower or phone. A nurse comes to them, and they keep their place in line.

HOW THE CONCEPT WORKS IN DETAIL

What's on the screen, and why

The tower and the phone use the same sections, in an order that suits each one. On the tower, the wait list sits at eye level for the whole room. On the phone, the patient's own ticket and the help button come first, within reach of their thumb.



Kiosk screen (one side of the tower)

1 Live status bar
Shows the clock and when the data last updated, so people can trust it's current.
Insight: the stress comes from not knowing what's going on.

2 Wait list, tap for the reason
Each ticket shows its status and estimated wait. Tapping a row shows why it changed. Rows show ticket numbers, never names, to protect privacy.
Insight: people want to know *why* they're waiting, not just how long (Israeli hospital study).

3 Hospital map
Shows "You are here," restrooms, food and exits. Closed areas are greyed out with a dashed border, so they read clearly without relying on color.
From Concept 2: helps patients and family move around without asking the desk.

4 Call for help
The largest button on both screens. It alerts the nurse station, and the patient keeps their place in line.
From Concept 3: a patient's condition can change, and some people are too sick or embarrassed to walk to the desk.

5 Videos, and taking it with you
Videos cover triage, staying hydrated and visitor rules, with captions on. A QR code opens the same screen on a phone. Languages can be switched.
From Concept 1: the patient's own phone is cheap and personal, and the texted link fixes the lost-ticket problem.

Built for a stressful room: large type and high contrast are readable from across the room. Touch targets are big enough for someone in pain. The screen cycles through all of its information on its own, so even people who never touch it are covered.

FROM EVALUATION TO CONCEPT

Why H.A.L.O. became the core, and what we borrowed

We compared our three concepts against our research insights and our classmates' feedback. No single concept covered everything, but each one's gap was another's strength.

	1 • QR ticket	2 • H.A.L.O. kiosk	3 • Scan from seat	H.A.L.O. system
Works without a phone	**No**	**Yes**	**Partly**: wall list only	**Yes**: tower
Shows estimated wait	**Yes**	**Yes**	**Yes**	**Yes**: both screens
Explains place in line / why	**No**	**Yes**: tap for reason	**Partly**: people ahead	**Yes**: both
Way to get help	**No**	**Yes**	**Yes**: notify nurse	**Yes**: from either screen
Map / wayfinding	**No**	**Yes**	**No**	**Yes**
Cost to the hospital	Low	High	Low	Start with one tower; phones are free
Peer concern	Paper ticket can get lost	Expensive to produce	Slow on bad Wi-Fi; one QR spot	Addressed below

Based on our concept papers and peer feedback. "Partly" means it works only in a limited way.

How each finding shaped the design

Stress comes mostly from **not knowing**. Telling ER patients their wait raised satisfaction 18% even when the wait was no shorter (DePaul).	→	Live wait times on every screen, with an "updated 12 sec ago" stamp.
Not everyone has a phone, a charged battery, or comfort with apps, especially older patients or people in pain.	→	The **tower is the core**. It cycles on its own, with nothing to download or scan.
People want to know **why**. Personal, updated information lowered frustration (Israeli hospital study).	→	Tap any ticket to see the reason its wait changed.
A patient's condition can **get worse while waiting**.	→	Call for help is on both screens, and the patient keeps their place.
Classmates rated H.A.L.O. strongest: "really good," "very thorough and easy to use," "self-explanatory." Their concern was cost.	→	The **phone mirror** stretches one tower further, so a hospital can start with one in its busiest waiting room.
Lost tickets (Concept 1) and slow Wi-Fi or a single QR spot (Concept 3).	→	A texted link as backup, a very light phone page, and QR codes on the ticket, the tower and the walls.

Sources

DePaul University: Announcing ER wait times improves patient satisfaction. business.depaul.edu

Israel National Institute for Health Policy Research: Personalized and dynamic information in the ER. israelhpr.org.il